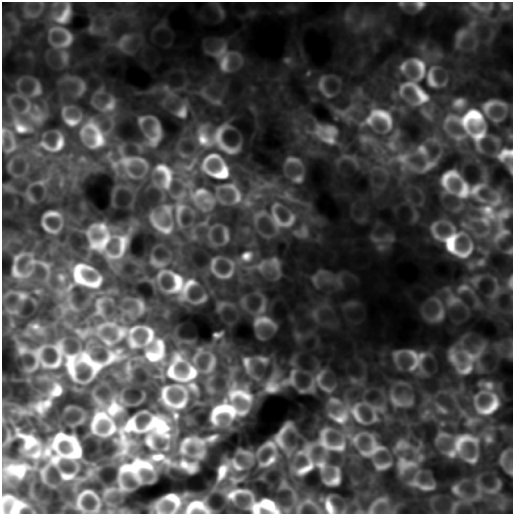
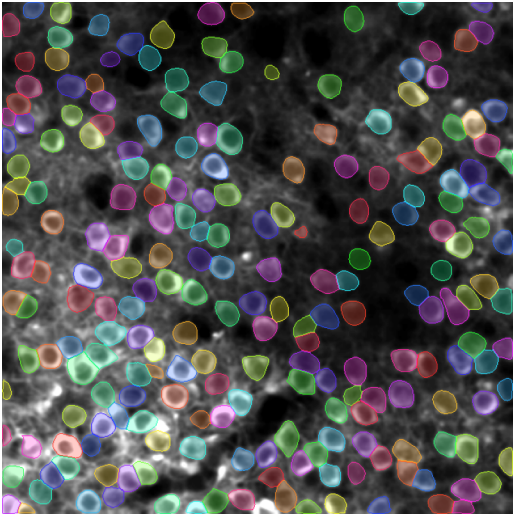
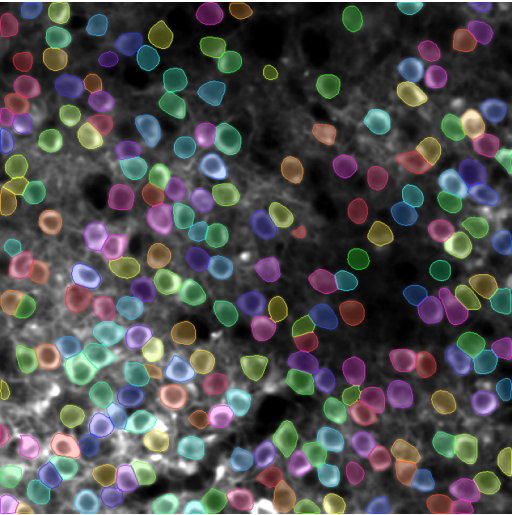
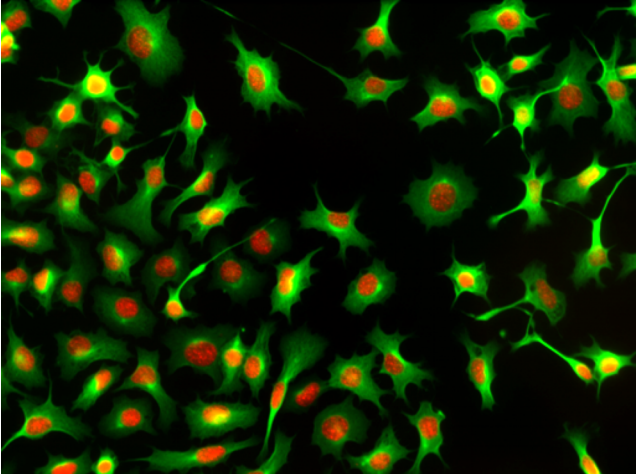
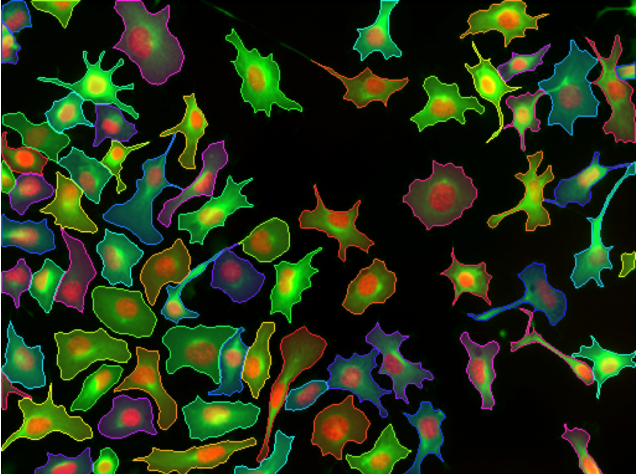
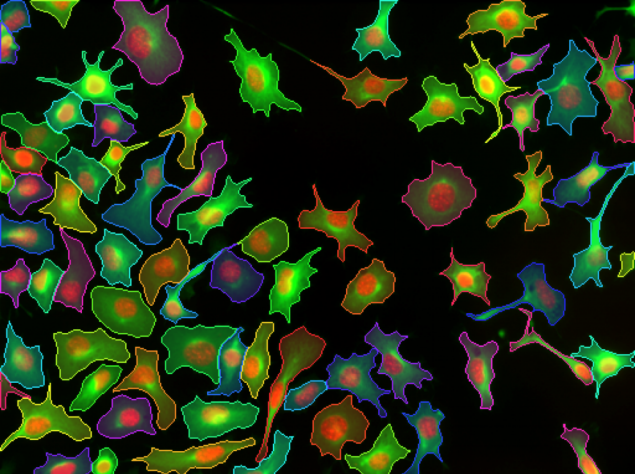
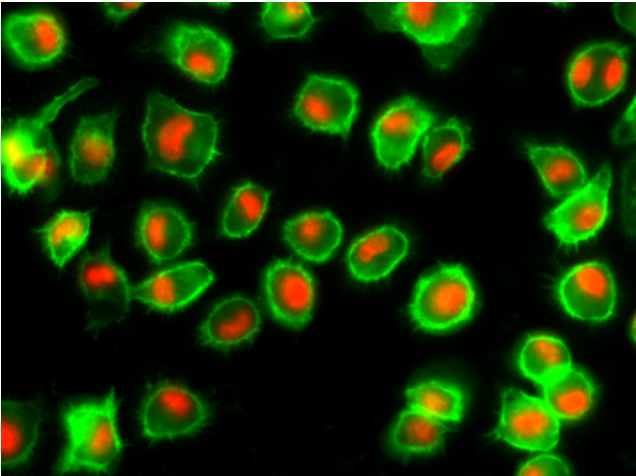
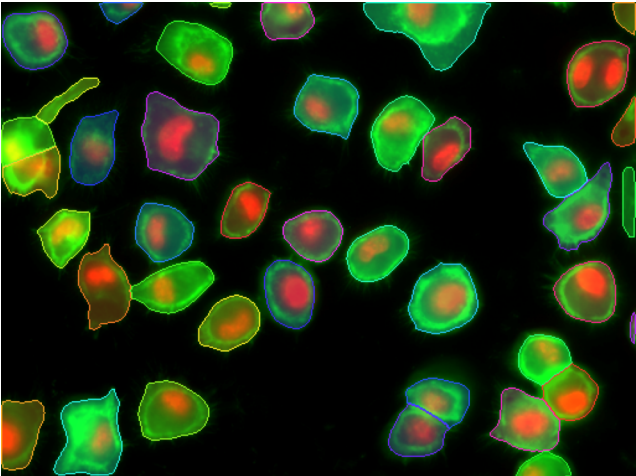
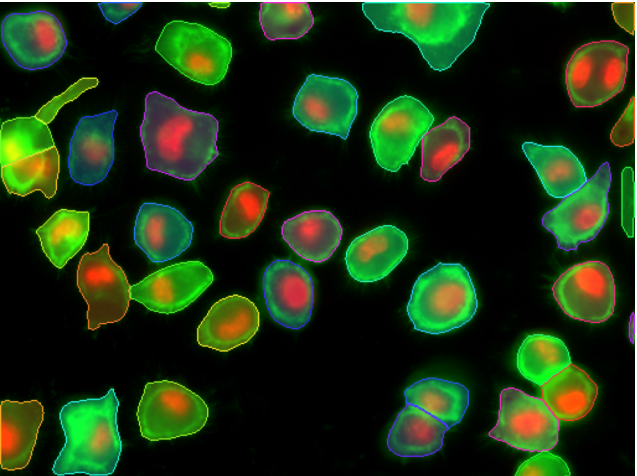
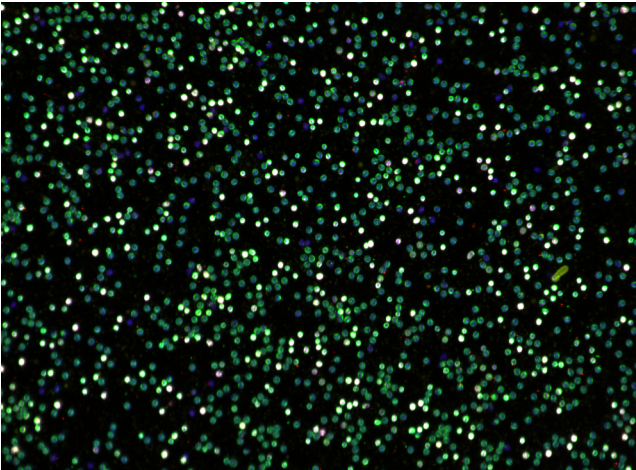
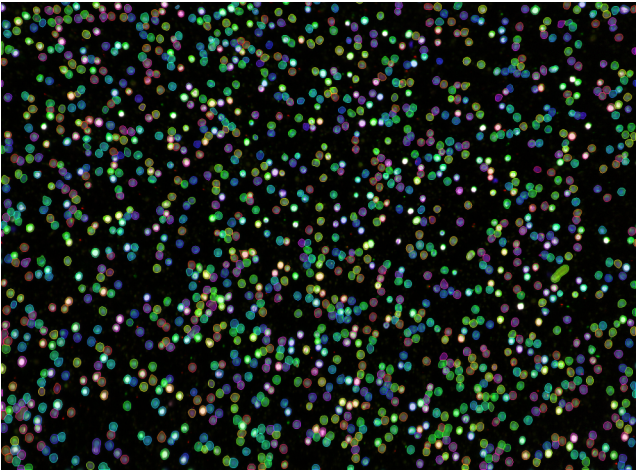
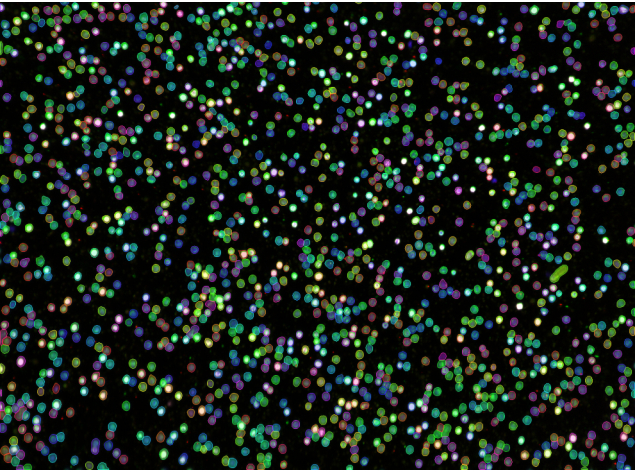


Cellpose 3 (cyto3): optimized inference preserves mask labels

2D images; cell diameter setting: 30 pixels; FP32; flow QC and iteration settings retained

	Input image	Upstream cell masks	Optimized cell masks	Label difference map
Grayscale PNG 677 x 677 pixels				0 changed label pixels 217 cells in each output Bitwise identical: yes 210.0 ms → 132.2 ms 1.59x faster 77.8 ms saved
RGB PNG 383 x 512 pixels				0 changed label pixels 75 cells in each output Bitwise identical: yes 157.0 ms → 122.3 ms 1.28x faster 34.7 ms saved
RGB TIFF 384 x 512 pixels				0 changed label pixels 41 cells in each output Bitwise identical: yes 147.8 ms → 108.6 ms 1.36x faster 39.2 ms saved
Microscopy A 1004 x 1362 pixels				0 changed label pixels 1523 cells in each output Bitwise identical: yes 823.3 ms → 396.2 ms 2.08x faster 427.1 ms saved

Matching colors denote the same cell label. Equality checks include every pixel, label ID, array shape and dtype.

Same label Different label

Times are medians from the five-repeat speed benchmark for each image; mask reproduction is a separate untimed check.